

Advanced Digital Design
130323
Xilinx Place & Route Lab (Assignment 8)
Date: Tuesday, January 21, 2025

Submission:	By not more than 2 people together
Deadline:	Motzei Shabat, February 8, 2025 (23:59) – via moodle
According to JCT's policy, if you violate any submission or deadline criteria you disqualify your assignment	

This exercise has two purposes:

- 1. Make you familiar with Vivado, Xilinx' place and route tool.**
- 2. Let you see and feel the silicon technology concepts that we studied**

As demonstrated in class, Vivado has a friendly GUI (Graphical User Interface), suitable for the new user, as well as internal TCL scripting capability, suitable for experts.

You are welcome to go through some dedicated training and tutorials for Vivado, available freely on YouTube (Google it).

You are also welcome to download Vivado and install on your computer

Notes:

1. The lab work is based on the 4-bit down counter, that was exercised in previous lab exercises.
2. You will go through a "Post Synthesis Project" using two files that were generated by Precision ("CNTR.v" and "CNTR.xdc") look into these files to understand what you are dealing with...

A. Lab Work – First Run

Take the netlist ("CNTR.v") and the constraints ("CNTR.xdc") and implement a new "Post Synthesis Project" targeted for an Artix-7 smallest device, in the smallest package, at the fastest speed grade (a "**-3**" speed grade).

1. Run the "Implementation" (don't forget to specify the top level module - "CNTR")
2. Open the implemented design and explore the device. Highlight each item of both the 'Nets' and the 'Leaf Cells' at the "Netlist" window to see these items in highlight at the "Device" window. (Zoom around the highlighted area in the device and take a screenshot of that area)
3. Go to the I/O ports tab
4. Highlight the ports listed at the I/O ports tab, select the "I/O Planning" at the top of the GUI. The You will see the I/Os. The "Device" window will now show the device' I/Os. Take a screenshot of the information. Make sure that you identify the location of each and every I/O pin associated with the design
5. Go to the "Reports" tab and collect the I/O Report, Utilization Report, and Timing Summary Report (include these reports in your final lab report)

6. Go to the “Timing” tab and make sure that all user timing constraints are met (take a screenshot of the information)
7. Highlight the “Project Manager” and the “Flow Navigator” and take a screenshot of the “Project Summary” window (For now, you can ignore the DRC errors. Find the DRC report by using the “Messages tab” and Include the “DRC report” in your report. What are the warnings all about?)

B. Lab Work – Second Run

Take the netlist (“CNTR.v”) and the constraints (“CNTR.xdc”) and implement a new “Post Synthesis Project” targeted for an Artix-7 smallest device, in the smallest package, at the slowest speed grade (a “-1” speed grade).

- Run the implementation flow – you should see timing violations. Find the information about these violations and include them in your report (explain why are there such violations)

C. Lab Report

Your lab report has to include the screenshots, reports and explanations that were requested within the above description of the lab.